



**UNIVERSITI MALAYSIA PAHANG
AL-SULTAN ABDULLAH**

FINAL ASSESSMENT

COURSE	:	CLOUD COMPUTING TECHNOLOGY
COURSE CODE	:	BCN3243
COURSE COORDINATOR	:	DR. ABDULLAH FAIRUZULLAH BIN AHMAD TAJUDDIN
DATE	:	30 DECEMBER 2025 TO 13 JANUARY 2026
DURATION	:	2 WEEKS

INSTRUCTIONS TO CANDIDATES:

1. This examination paper consists of **TWO (2)** questions. Answer **ALL** questions.
2. Candidates are not allowed to bring any materials other than those allowed by the invigilator into the examination room.
3. For **QUESTION 1-2**, please refer to the attached rubric (Appendix 1).

FINAL ASSESSMENT REQUIREMENTS:

1. Free Credit Cloud Account (Huawei Cloud, Microsoft Azure, Google Cloud, Oracle Cloud)

APPENDIX:

1. Appendix 1 – Rubric evaluation

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of **TEN (10)** printed pages including the front page.

CASE STUDY

Edu Fairuzullah Sdn Bhd is a growing company startup that provides online learning and virtual classroom services for schools, universities, and corporate clients. The company plans to launch a cloud-based Learning Management System (LMS) that allows educators to create courses, upload learning materials, host live sessions, and manage student assessments.

Edu Fairuzullah's system will have two main user groups:

- 1) Educators: Instructors who create and manage courses, upload resources, and host virtual classes.
- 2) Learners: Students who register for courses, access learning materials, participate in live classes, and complete assessments.

Edu Fairuzullah has a small IT team experienced in web development but limited in managing scalable, secure, and distributed cloud infrastructure. They plan to deploy their platform on a public cloud provider (such as Microsoft Azure, AWS, Google Cloud Platform, Huawei Cloud, Oracle Cloud, or Alibaba Cloud) to take advantage of cloud scalability, flexibility, and pay-as-you-go pricing.

The Business and Technical Requirements ensure that the LMS is secure, scalable, high-performing, cost-efficient, and easy to manage for all users

a. Functionality

- A public-facing website to browse available courses and register as a learner or educator.
- A secure educator portal for course management and virtual class hosting.
- A secure learner portal for content access and participation.
- Storage for course materials, videos, and user data.

b. Performance

- Fast, responsive access for users across regions.
- Efficient handling of traffic spikes during course enrollments or live sessions.

c. Availability

- High availability and minimal downtime, especially during live teaching sessions.
- Scalability
- Automatically scale based on user demand and class load.

d. Security

- Enforce user authentication and role-based authorization (Educator vs. Learner).
- Protect all sensitive data (in transit and at rest).
- Mitigate common web vulnerabilities (SQL injection, DDoS, cross-site scripting).
- Comply with data privacy laws (e.g., GDPR, PDPA).

e. Management

- Reduce manual administration for the small IT team.
- Enable centralized monitoring, logging, and alerting for system performance.
- Cost
- Operate under a pay-as-you-go model.
- Optimize costs while maintaining performance and reliability.

QUESTION 1 [80 MARKS]

From the given case study, as a Cloud Engineer, you are required to propose a comprehensive cloud solution using only one provider of the six public cloud computing platforms (e.g., AWS, Azure, Google Cloud, Huawei Cloud, Oracle Cloud, or Alibaba Cloud). Please answer the following questions accordingly..

(a) Compute Service Selection

- i. Identify and describe at least **TWO (2)** suitable compute services offered by *any* major public cloud provider that could host the Edu Fairuzullah LMS components (web app, backend APIs, live sessions). For each service, state its service model (IaaS, PaaS, or SaaS) and explain why it fits Edu Fairuzullah's needs.

[8 Marks]

- ii. Choose the most appropriate compute service based on Edu Fairuzullah's team expertise, scalability, and management constraints. Justify your choice with reference to cloud design principles and the case study requirements.

[8 Marks]**(b) Database Service Selection**

- i. Identify and describe at least **TWO (2)** suitable database services (relational or NoSQL) from any cloud provider to store course content, user accounts, and learning records. Discuss each service's characteristics and suitability for Edu Fairuzullah's data patterns and access needs.

[8 Marks]

- ii. Select the most appropriate database service for Edu Fairuzullah. Justify your selection based on scalability, cost, data consistency, and management requirements.

[8 Marks]

(c) Identity and Access Management (IAM)

- i. Explain which cloud IAM service could be used to manage authentication and authorization for learners and educators.
[8 Marks]
- ii. Describe how this service provides secure access control and integrates with the LMS, referring to relevant cloud security principles (least privilege, identity federation, zero trust).
[10 Marks]

(d) Networking and Security

- i. Identify and describe **TWO (2)** cloud networking or security services/concepts (e.g., firewalls, VPCs, WAFs, encryption, CDN) that are essential to protect Edu Fairuzullah's LMS and its users' data.
[6 Marks]
- ii. Explain how each selected service supports Edu Fairuzullah's security posture, referencing principles such as defense-in-depth, network isolation, and secure connectivity.
[10 Marks]

(e) Management, Monitoring, and Cost

- i. Identify and describe **ONE (1)** cloud service or tool that helps Edu Fairuzullah monitor system performance and health. Explain its importance and how it would be implemented for the LMS.
[6 Marks]
- ii. Mention **ONE (1)** cost management or optimization tool available from a public cloud provider that EduWave could use to estimate and control costs. Justify your answer based on the startup's operational goals and scalability model.
[10 Marks]

QUESTION 2 [20 MARKS]

Based on your proposed architecture in **Question 1(a) Compute Service Selection** and **Question 1(b) Database Service Selection**, design and develop a prototype Learning Management System (LMS) demonstrating the fundamental components of Edu Fairuzullah's solution. Your prototype should

- Include basic CRUD operations (Create, Read, Update, Delete) for course and user data. **[10 Marks]**
- Implement authentication and authorization and Support two user roles: Educator and Learner. **[5 Marks]**
- Deployable on any public cloud provider of your choice **[5 Marks]**

Refer to Appendix A for evaluation criteria and demonstration requirements.

END OF QUESTION PAPER

APPENDIX 1 – RUBRICS EVALUATION

QUESTION 1 [80 MARKS]

(a): Compute Service Selection (16 marks)

i. Identify & describe two compute services, model and justification (8 marks)

Level	Description	Marks
Excellent	Two compute services clearly identified; correct service models (IaaS/PaaS/SaaS); explanations show strong alignment to LMS needs (scalability, hosting models, traffic load).	7–8
Good	Two compute services described with mostly correct details; justification partially aligned to the case study.	5–6
Satisfactory	Services identified but lacking clarity in model or justification.	3–4
Weak	Incorrect/unclear services; missing service model; poor alignment to requirements.	0–2

ii. Select the most appropriate compute service and justification (8 marks)

Level	Description	Marks
Excellent	Clear selection with strong justification using cloud principles (stateless design, scaling, managed service, team expertise).	7–8
Good	Clear choice but justification lacks depth or incomplete principles.	5–6
Satisfactory	Choice stated with minimal justification.	3–4
Weak	No clear choice; weak or irrelevant justification.	0–2

(b): Database Service Selection (16 marks)

i. Identify & describe two database services + suitability (8 marks)

Level	Description	Marks
Excellent	Two database services (SQL/NoSQL) well explained; characteristics match Edu Fairuzullah's data patterns; strong reasoning.	7–8
Good	Two services identified with reasonable description.	5–6
Satisfactory	Services listed but explanations weak or generic.	3–4
Weak	Incorrect choices or poorly explained suitability.	0–2

ii. Select most appropriate DB service and justification (8 marks)

Level	Description	Marks
Excellent	Choice justified with strong arguments on consistency, scaling, cost, and managed operations.	7–8
Good	Good justification but minor gaps.	5–6
Satisfactory	Limited justification; general statements.	3–4
Weak	Wrong selection or very poor reasoning.	0–2

(c). IAM (18 marks)**i. Identify appropriate IAM service (8 marks)**

Level	Description	Marks
Excellent	Correct IAM service identified; clearly explained for both Educator & Learner roles.	7–8
Good	Mostly correct IAM usage with minor gaps.	5–6
Satisfactory	IAM named but not well explained.	3–4
Weak	Incorrect or vague IAM.	0–2

i. Integration & security principles (10 marks)

Level	Description	Marks
Excellent	Clear explanation referencing Zero Trust, least privilege, identity federation; strong LMS integration reasoning.	9–10
Good	Good security reasoning but not comprehensive.	7–8
Satisfactory	Generic discussion of access control.	4–6
Weak	Incorrect or superficial explanation.	0–3

(d). Networking & Security (16 marks)**i. Identify & describe two services (6 marks)**

Level	Description	Marks
Excellent	Two correct services (e.g., WAF, CDN, VPC, firewall, encryption); strong relevance.	5–6
Good	Services identified but lightly explained.	3–4
Weak	Incomplete or incorrect services.	0–2

ii. Security justification (10 marks)

Level	Description	Marks
Excellent	Clear role of each service in defense-in-depth, network isolation, secure connectivity.	9–10
Good	Mostly correct but not fully aligned with principles.	7–8
Satisfactory	Generic or basic reasoning.	4–6
Weak	Poor or incorrect explanation.	0–3

(e) Monitoring & Cost Management (14 marks)**i. Monitoring tool (6 marks)**

Level	Description	Marks
Excellent	Correct cloud monitoring tool; clear implementation for LMS performance & health.	5–6
Good	Good description with minor missing details.	3–4
Weak	Generic, incomplete, or incorrect.	0–2

ii. Cost optimization tool (10 marks)

Level	Description	Marks
Excellent	Clear tool identified (e.g., AWS Cost Explorer, Azure Cost Management); strong justification tied to startup cost model.	9–10
Good	Good justification.	7–8
Satisfactory	Generic statements.	4–6
Weak	No cost tool identified or incorrect.	0–3

QUESTION 2 (20 MARKS)**Assessment Scope:**

Prototype LMS including roles, CRUD operations, cloud deployment, database integration, authentication, and architecture alignment with Q1.

Functional Requirements (10 marks)

Level	Description	Marks
Excellent	Fully working prototype; both roles (Educator & Learner); all CRUD operations implemented; DB integrated; clean structure.	9–10
Good	Most features implemented but minor missing details.	7–8
Satisfactory	Partial implementation (e.g., only some CRUD).	4–6
Weak	Very minimal or non-functional implementation.	0–3

Authentication & Authorization (5 marks)

Level	Description	Marks
Excellent	Secure login system; role-based access control; cloud IAM/custom auth fully integrated.	5
Good	Auth implemented but limited RBAC or incomplete integration.	3–4
Weak	Minimal or faulty authentication.	0–2

Cloud Deployment (5 marks)

Level	Description	Marks
Excellent	Successfully deployable on chosen cloud provider; clear deployment structure (compute + database).	5
Good	Partially deployable but missing steps/details.	3–4
Weak	No clear deployment capability.	0–2